Hans Hunter: catching medical image classifiers that cheat, without knowing how they cheat

Shortcut forensics by reverse-engineering the cheapest edit that flips a diagnosis, tested on 210 models with planted shortcuts across skin, chest X-ray and breast ultrasound images

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Abstract. Medical image classifiers often learn shortcuts, cues such as a hospital's corner marker, a ruler or a scanner's colour cast that happen to accompany a diagnosis in the training data. The model's test score does not reveal them, and a shortcut model can look *better* than an honest one. We build a controlled benchmark of 210 small convolutional networks on three MedMNIST tasks. Each network was trained while one of five synthetic artifacts (including a ±3/255 pattern that is invisible to a person) accompanied the disease with probability p from 0.5 (unrelated) to 1.0 (always). We then audit each model without telling the auditor which artifact, if any, was planted. The audit searches for the cheapest universal edit (a small patch, a faint additive pattern, or a colour change) that turns healthy scans into "disease" predictions. Four findings stand out. (1) Reliance on the artifact switches on at about $p \approx 0.8$. At $p = 0.9$, a model can keep a new-hospital AUC of 0.94 while flagging 40% of healthy patients whose scan happens to carry the marker, so AUC hides the harm. (2) The cost of the cheapest flipping patch separates shortcut models from honest ones with AUROC 0.95 on breast ultrasound, 0.77 on skin and only 0.64 on chest X-ray. Honest X-ray models are themselves cheap to flip with disease-like edits, which confounds the cost signal. Even so, the recovered triggers visibly *point at the artifact* and reproduce the invisible pattern, which saliency maps cannot show. (3) Turning the trigger into a direction in feature space finds which images carry the shortcut (within-class AUROC 0.91–0.97 on skin and breast). It is only slightly better than ranking images by the model's own score, a baseline we did not expect to be so strong. (4) Fine-tuning with the recovered triggers ("trigger scrubbing") repairs global shortcuts. For the invisible pattern on breast ultrasound, new-hospital AUC goes from 0.72 to 0.87 against 0.87 for a model trained without the shortcut, and false alarms caused by the pattern go from 42% to 0%. It fails for localized markers. Reweighting with human-confirmed groups repairs most shortcuts; with automatically inferred groups it helps little. No method rescues a model trained at $p = 1.0$. Applied to the real datasets, the same tools find no compact shortcut. They do find that 16.7% of BreastMNIST test images have a near-identical copy in its training set, two of them with the opposite label. Code, all 210 models' results, and an in-browser demo are public.

1 · Introduction

A deep network trained to spot pneumonia in chest X-rays was found to be partly identifying *which hospital* an X-ray came from, using markers burned into the image corners, because pneumonia was far more common at one hospital [Zech 2018]. COVID-19 detectors learned laterality tokens and image-source differences [DeGrave 2021]. Dermatology classifiers rated

benign lesions as more likely malignant when a surgical skin marking was visible [Winkler 2019]. Such shortcuts [Geirhos 2020] are named after *Clever Hans*, the horse that appeared to do arithmetic but was reading its trainer's posture. They are dangerous for a specific reason: on data from the same source, the shortcut *helps* the score, so standard validation rewards it.

Most existing tools ask the auditor to *name* the suspected confounder first (test with and without rulers), or to read saliency maps image by image. We ask a different question: **given only a trained model and some healthy images, can an auditor find out that the model has a shortcut, what the shortcut looks like, which training images carry it, and repair the model?** We borrow the idea behind backdoor detection [Neural Cleanse, Wang 2019]. If a small, universal edit is enough to flip many healthy images to "disease", that edit is a candidate shortcut. A planted backdoor and an accidental shortcut are the same object seen from different sides.

Real shortcuts are not labelled, so we cannot measure how well an audit works on them. We therefore plant artifacts with known strength into real medical images, train many models, and grade the audit against the planted truth. That turns a qualitative worry into a benchmark, and it shows clearly where the method fails.

2 · Setup

Data. Three binary tasks from MedMNIST v2 [Yang 2023] at 64×64: *DermaMNIST* (HAM10000 dermatoscopy; malignant = melanoma, basal cell carcinoma or actinic keratosis; 7,007 / 1,003 / 2,005 train/val/test), *PneumoniaMNIST* (pediatric chest X-ray; 4,708 / 524 / 624) and *BreastMNIST* (breast ultrasound, malignant vs. not; 546 / 78 / 156, downsampled from 128 px). Together the files are 131 MB.

Artifacts. Five artifacts, stamped with small random variation in position and intensity (Figure 1): a corner *tag* (an "R" glyph, as in hospital laterality markers), a *ruler* along the bottom edge, an *ink* ellipse around the centre (a surgical skin marking), a global *tint* (one scanner's colour cast), and an *invisible* tiled 8×8 pattern of amplitude ±3/255. In training, an image carries the artifact with probability p if diseased and $1-p$ if healthy, for $p \in \{0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0\}$. At $p = 0.5$ the artifact is independent of the label, so that model is the *honest* reference.

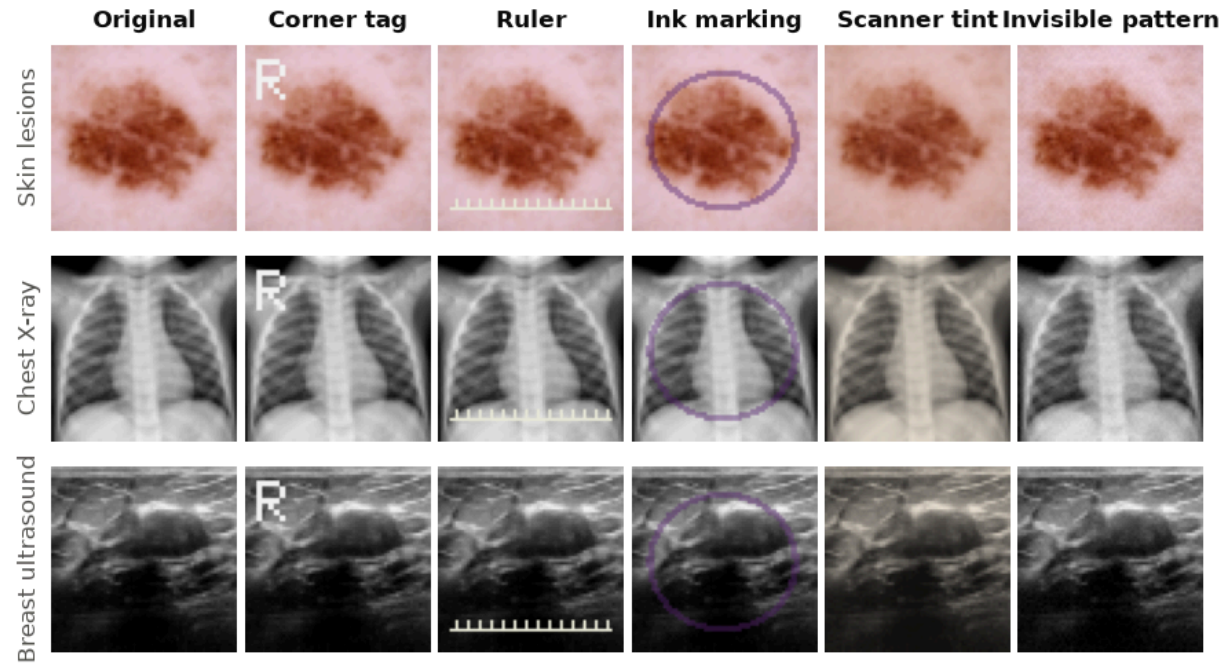


Figure 1. The five planted artifacts on one image from each dataset. The invisible pattern (rightmost) changes pixels by at most 3/255.

Models. A 1.27M-parameter ResNet (four stages, 32–128 channels), trained with AdamW and a one-cycle schedule, flip and ±4 px shift augmentation, for 20 / 25 / 60 epochs (skin / X-ray / breast), two seeds each: 3 × 5 × 7 × 2 = 210 models. Everything runs on one laptop GPU.

Evaluation. Every model is scored at three "hospitals" built from the same test images: *source* (artifact correlated as in training), *new* (artifact present at random, independent of the label) and *flipped* (the correlation reversed). Because AUC can hide a shortcut, we also report **false alarms**: the share of healthy test patients predicted diseased once the artifact is added, minus the share without it.

3 · When does a model start cheating, and what does it cost?

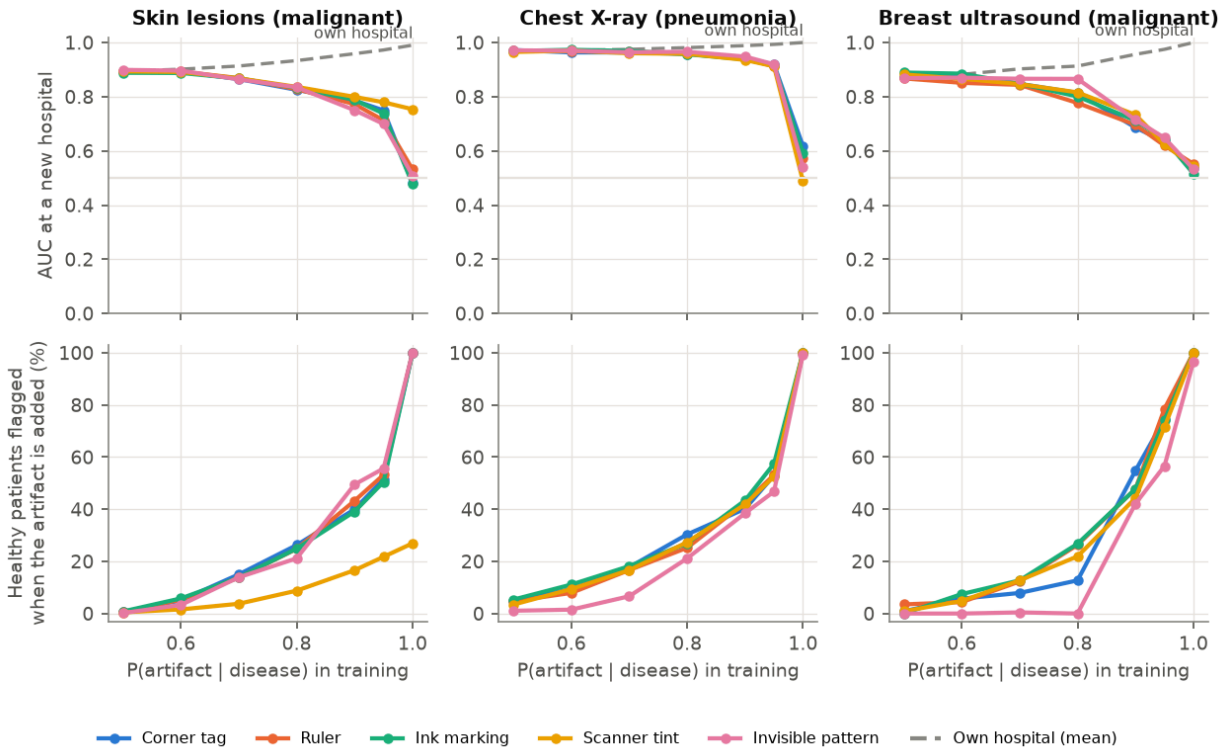


Figure 2. Top: AUC at a new hospital (solid, by artifact) and at the training hospital (dashed). Bottom: healthy patients flagged as diseased when the artifact is added. Means over two seeds.

Figure 2 is the phase diagram. As p grows, source AUC rises (to 1.00 at $p = 1$) while new-hospital AUC falls. Reliance turns on gradually up to $p \approx 0.7$ and steeply after 0.8. At $p = 1$ the model ignores the anatomy and new-hospital AUC is at chance (0.48–0.62 for every visible artifact). Three details matter:

- **AUC can hide the harm.** On chest X-rays at $p = 0.9$, the new-hospital AUC is 0.94 against 0.97 for the honest model. That is a small drop that would pass most reviews. Yet adding the marker makes the model flag **40%** of healthy children as having pneumonia. The shortcut adds to the evidence from the lungs instead of replacing it, so the ranking of patients within a hospital survives while the operating point moves.
- **The invisible pattern is learned as readily as visible markers,** but on breast ultrasound only past a threshold: new-hospital AUC stays at 0.87 up to $p = 0.8$ and then drops to 0.71 at 0.9.
- **Skin models partly resist the colour cast.** Even at $p = 1$, the tinted skin model keeps a new-hospital AUC of 0.75 and flags only 27% of healthy patients. A plausible reason is that lesion colour is itself diagnostic and varies naturally across patients, so a colour cast competes with real signal.

4 · The audit: reverse-engineering the cheapest flip

For a model f and 64 healthy held-out images x from the source hospital, we search for a universal edit that makes ≥80% of them predicted diseased, at minimum cost. We use three families, each

optimised with Adam and a Lagrangian weight on the cost that is adjusted automatically [Wang 2019]:

- **patch**: $x' = (1-m)\odot x + m\odot P$; cost = $\|m\|_1$ in pixels;
- **additive**: $x' = \text{clip}(x + \delta)$; cost = RMS($\delta$) in 1/255 units;
- **colour**: per-channel gain and offset; cost = RMS pixel change.

The auditor is not told the artifact, its location, or p. One audit takes about 18 s per model on the GPU. The idea is simple. An honest model can be flipped only by edits that look like disease, which costs many pixels. A shortcut model has a cheap key.

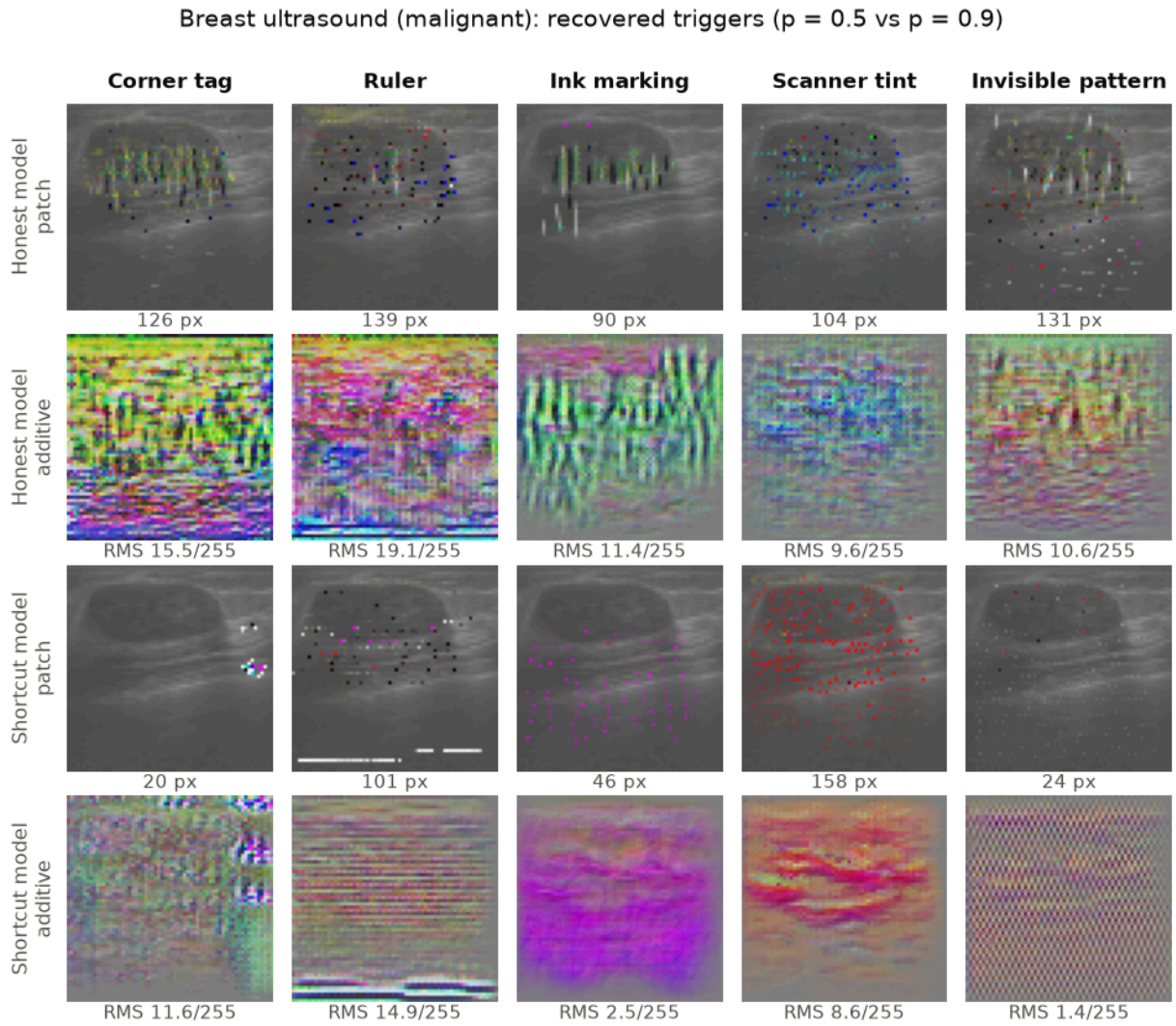


Figure 3. Triggers recovered from breast ultrasound models. Rows 1–2: honest models (p = 0.5); rows 3–4: shortcut models (p = 0.9). Patch triggers are drawn over a dimmed image, with their size in pixels below each one. Additive triggers are shown amplified, with their RMS below. For honest models the edit is large and diffuse. For shortcut models it points at the planted artifact: a small cluster at the image edge where the tag sits (20 px; the edge alternates because of flip augmentation), white lines along the bottom where the ruler sits, a reddish colour cast for the tint, and a fine periodic texture for the invisible pattern (additive RMS 1.4/255).

Detection. Figure 4 shows the cost of the cheapest flipping patch as a function of p. Using that cost to separate honest models (p = 0.5) from strong-shortcut models ($p \geq 0.9$), the AUROC is

0.945 on breast ultrasound (median 151 px vs. 30 px), **0.765** on skin (34 vs. 12 px) and **0.635** on chest X-ray (24 vs. 19 px). The additive cost is weaker (0.83 / 0.61 / 0.60). The X-ray result is the important negative one. Honest pneumonia models are *already* cheap to flip, because 20–30 pixels of disease-like opacity anywhere in the lung is enough. A low cost therefore does not prove a shortcut. The cost is a screening statistic, and reading the recovered trigger is what settles the question.

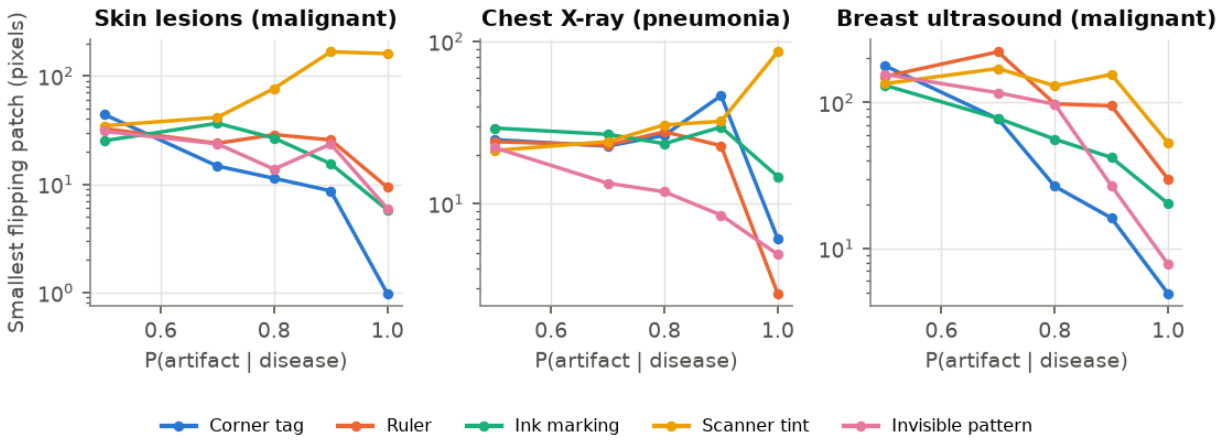


Figure 4. Size of the smallest universal patch that flips 80% of healthy images (log scale). On breast ultrasound, cost falls by up to 30× as the shortcut strengthens; on chest X-ray, honest models start low. Skin models with the colour cast get *harder* to flip with a patch (35 → 167 px), because the shortcut they learned is global, not local.

Do the triggers find the artifact? For the three localized artifacts, we measure what share of the recovered patch mask falls on the artifact's region. For the tag on skin and breast images, that share rises from about chance at $p = 0.5$ (e.g. 0.16 on breast, where the region covers 0.21 of the image) to 0.5–0.6 at $p \geq 0.8$; on chest X-rays it reaches 0.5 only at $p = 1$. For the ruler at $p = 1$, the patch lies almost entirely on the ruler (0.99 on skin and X-ray).

Comparison with saliency maps

The usual way to look for shortcuts is a saliency map. Figure 5 compares Grad-CAM [Selvaraju 2017] with the trigger on the three visible artifacts. Grad-CAM *does* move toward visible artifacts as p grows: on breast ultrasound the share of Grad-CAM mass on the corner tag goes from 0.03 to 0.33, against an area share of 0.06. It is a fair baseline that we should not dismiss. It has two limits that the trigger does not. First, it explains one image at a time, and an auditor must read many maps. Second, it can only point at *where*, not *what*. A heat map over the whole image says nothing about a 3/255 texture or a colour cast, while the additive trigger reproduces the texture itself (for the invisible pattern on breast, RMS falls from 17/255 to 0.7/255 as p goes 0.5 → 1). The recovered patch is also usually more concentrated on the artifact than Grad-CAM (dashed vs. solid lines).

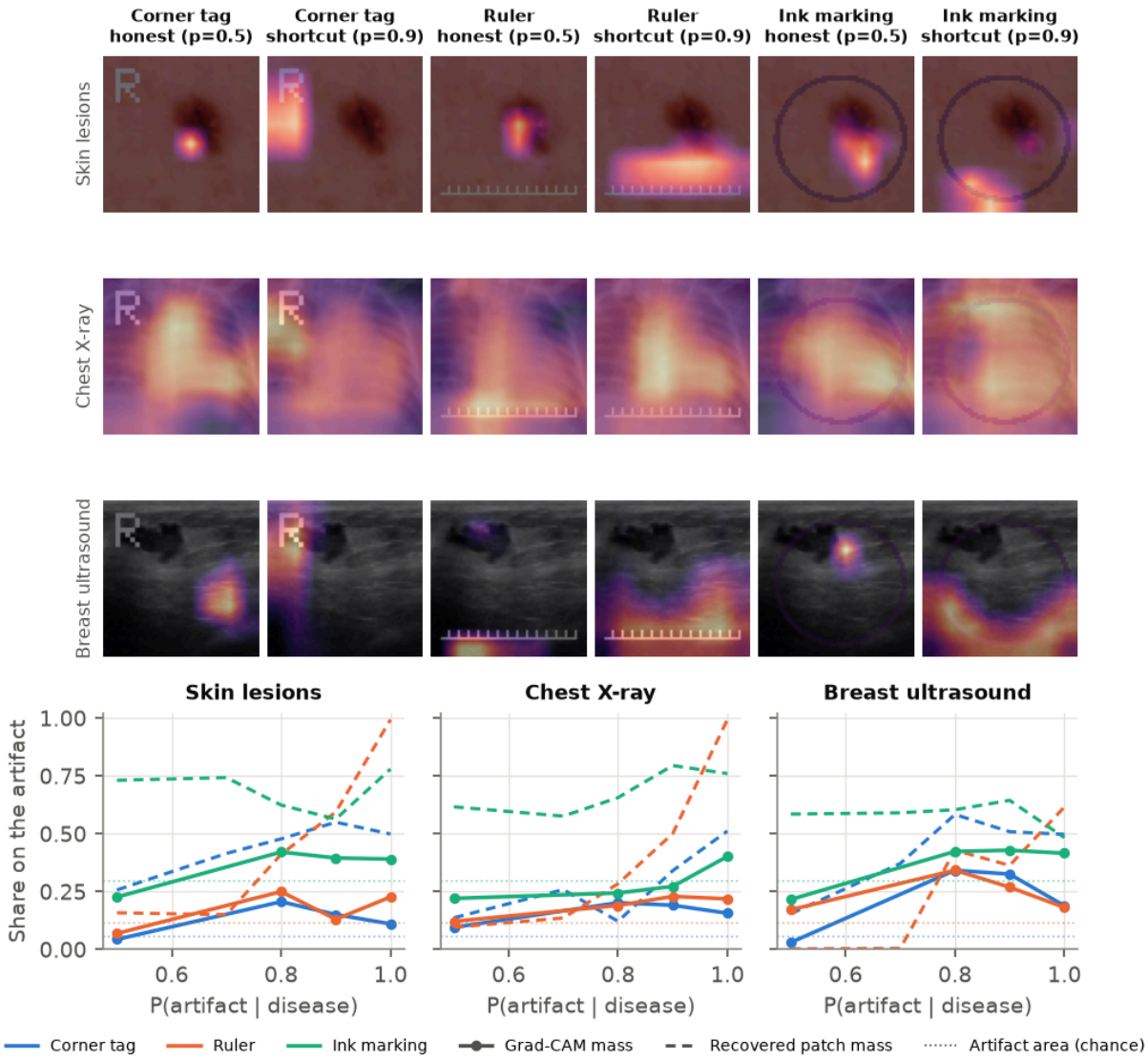


Figure 5. Top: Grad-CAM on one healthy image with each visible artifact, for honest (p = 0.5) and shortcut (p = 0.9) models. Bottom: share of Grad-CAM mass (solid) and of the recovered patch mask (dashed) on the artifact region; dotted lines show the share expected by chance (the artifact's area).

5 · Which training images carry the shortcut?

To repair a model, or to warn the people who collected the data, the auditor needs to know which images carry the cue. We apply the recovered patch and additive triggers to the audit images and average the resulting change in the network's 128-d penultimate features. This gives a *trigger direction* t. Each image is scored by its projection on t after subtracting its class mean, and we measure, within each class, how well the score separates images with and without the planted artifact.

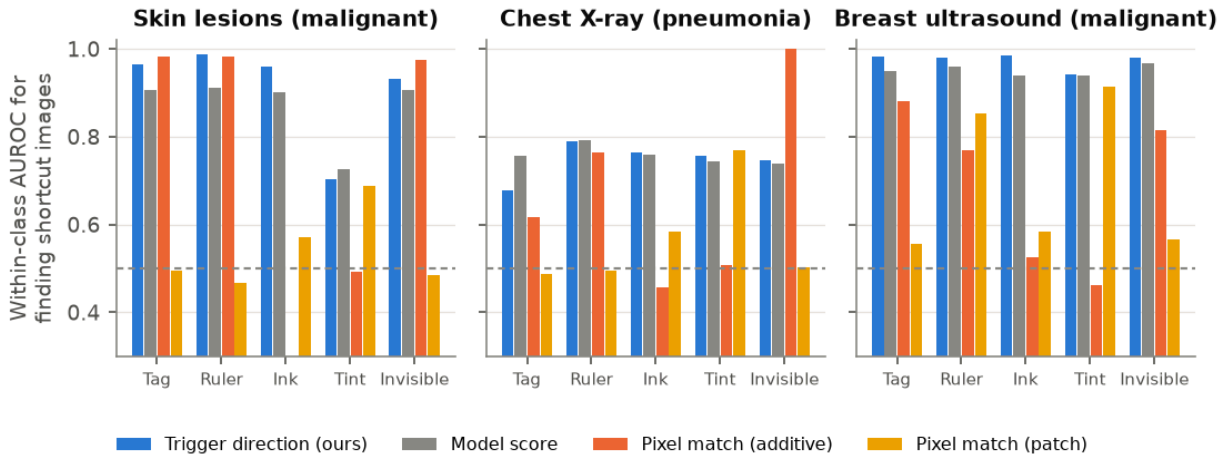


Figure 6. Within-class AUROC for finding images that carry the artifact (held-out source-hospital images, p = 0.9). Blue: the trigger direction; grey: the model's own output; orange/yellow: matching the trigger in pixel space.

Table 1. Within-class AUROC for finding shortcut-carrying images, averaged over artifacts and seeds.

Dataset	p = 0.5	0.7	0.8	0.9	model score, 0.9	best pixel match, 0.9
Skin	0.51	0.71	0.81	0.91	0.87	0.73
Chest X-ray	0.51	0.60	0.65	0.75	0.76	0.67
Breast ultrasound	0.51	0.68	0.77	0.97	0.95	0.69

The direction finds the affected images well when the shortcut is strong (0.91–0.97 on skin and breast). It is at chance for honest models, as it should be. Matching the trigger in pixel space is unreliable: it works for some artifacts and fails completely for others. The surprise is the **model-score baseline**. Simply ranking healthy images by how diseased the model thinks they are gives nearly the same result (0.87 / 0.76 / 0.95). Once a shortcut is strong, it is the main reason a healthy image scores high. The trigger direction adds only 0.02–0.04 on top. We report this plainly, because a method that only beats a straw-man baseline would be easy to oversell. The direction should matter more when a model has several shortcuts or when disease signal dominates the score, but we have not tested that here. At p = 1 the question is undefined (every healthy image is artifact-free).

6 · Repairing the model

We compare five repairs on the p = 0.9 models, scored on new-hospital AUC (Figure 7, Table 2):

- **Trigger scrubbing**: fine-tune the model while stamping the recovered triggers onto random training images (both classes), so the trigger stops carrying information; two rounds of audit and fine-tuning. A fine-tune without triggers is the control.
- **Reweighting with inferred groups**: split each class at the Otsu threshold of the trigger-direction score, then retrain with the four (class × group) cells weighted equally, in the style of group-balanced retraining [Sagawa 2020; Liu 2021].

- **Reweighting with confirmed groups**: the same, but with the true artifact labels. This stands for an auditor who has seen the trigger, recognized it ("the R marker") and labelled the images that carry it.
- **Projecting out** the group direction before the classifier head, with inferred or confirmed groups, with no retraining.
- **Oracle**: a model trained with the artifact unrelated to the label.

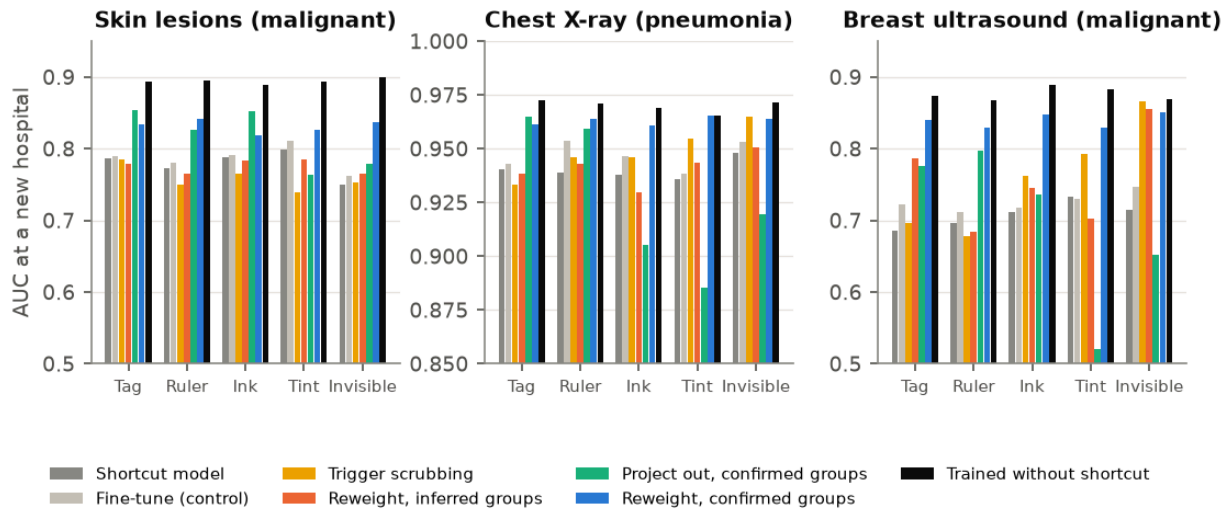


Figure 7. New-hospital AUC of $p = 0.9$ models after each repair (two seeds).

Table 2. New-hospital AUC at $p = 0.9$, mean over artifacts and seeds. Scrubbing is reported per artifact in the text.

Dataset	shortcut	scrub	reweight, inferred	project, confirmed	reweight, confirmed	oracle
Skin	0.78	0.76	0.78	0.82	0.83	0.89
Chest X-ray	0.94	0.95	0.94	0.93	0.96	0.97
Breast ultrasound	0.71	0.76	0.76	0.70	0.84	0.88

Scrubbing fixes global shortcuts and not local ones. On breast ultrasound with the invisible pattern, scrubbing restores new-hospital AUC from 0.72 to **0.87** (oracle 0.87) and cuts false alarms from the pattern from 42% to **0%**. For the colour cast, AUC goes 0.73 → 0.79 and false alarms 44% → 9%. On chest X-ray it removes most false alarms from the pattern (38% → 9%) and the cast (42% → 25%). For the tag, ruler and ink, and on skin in general, it does nothing beyond the fine-tuning control. Our reading is that a global shortcut has essentially one form, so the trigger captures it completely and stamping it teaches the model to ignore it. A local marker is recognized by features that a single recovered patch only partly covers, so the model learns to ignore that particular patch and keeps using the marker.

Groups are the bottleneck. Reweighting with confirmed groups recovers most of the gap to the oracle everywhere (breast 0.71 → 0.84 vs. 0.88; skin 0.78 → 0.83 vs. 0.89). With inferred groups the gain mostly disappears: the Otsu split agrees with the truth only 73% of the time on average,

and balancing noisy groups does little. The exceptions are breast tag (0.69 → 0.79) and breast pattern (0.72 → 0.86). Projecting out an *inferred* direction is harmful (it removes disease signal, mean AUC falling to 0.59–0.80) and we do not recommend it. A practical workflow follows. The audit finds and draws the shortcut, a person confirms it and labels where it appears, and standard group-balanced retraining does the repair.

Some damage cannot be repaired. At $p = 1$, neither scrubbing nor fine-tuning gets above 0.68 new-hospital AUC on any dataset (we ran the reweighting repairs only at $p \leq 0.9$; with no artifact-free diseased images, reweighting has nothing to balance). When every diseased training image has the artifact and no healthy one does, the data simply does not say whether the disease or the artifact is the cause. The audit still detects these models best of all (a 1-pixel trigger flips the skin tag model). What it can tell the user in that case is that new data must be collected.

7 · Auditing the real datasets

We ran the audit on three clean models per real dataset (no planted artifact). The cheapest edits that push a healthy image to "disease" cost 12 px (skin), 17 px (X-ray) and 208 px (breast), the same range as our honest synthetic models. The triggers look like small lesion-like or opacity-like edits rather than a recurring marker. **We find no evidence of a compact shortcut** in these 64-pixel versions of the datasets, which is not the same as showing there is none (Section 9).

A side check found something else. Comparing every pair of images across splits by the correlation of 32×32 mean-centred thumbnails, we find that **26 of 156 BreastMNIST test images (16.7%)** have a near-identical copy ($r > 0.99$) in the training set, plus 18 of 78 validation images. In two of those pairs the copies have *opposite labels* (Figure 8). DermaMNIST has 20 such test images (1.0%); the classifiers get 19–20 of them right, against 85–86% accuracy on the rest. PneumoniaMNIST has none. These may be repeated frames of the same lesion rather than literal copies, but either way they leak training data into the test set and inflate the reported numbers for this widely used benchmark.

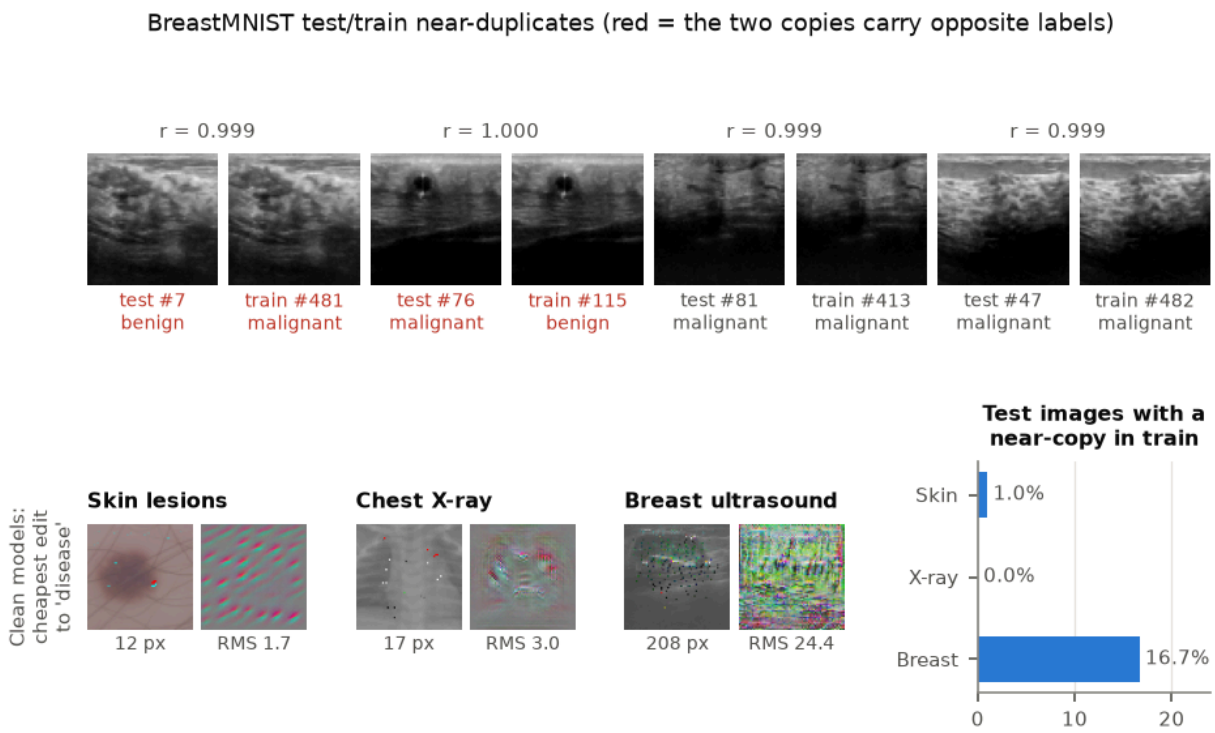


Figure 8. Top: BreastMNIST test images and their nearest training images; red labels mark pairs whose labels disagree. Bottom: triggers recovered from clean models on the real data (patch, additive), and the share of test images with a near-copy in train.

8 · Things that did not work

We list the negative results because they are useful to anyone trying this:

- **Sparse-autoencoder feature audit.** A TopK sparse autoencoder on the last feature map gives exact per-feature contributions to the logit. On shortcut models, however, the artifact did not separate into its own feature: it was mixed with disease features, and ranking features by contribution did not reliably point to it.
- **Within-class contrast images, bimodality and "label-blindness" statistics** of feature directions were noisier detectors than trigger cost.
- **Grouping by Gaussian mixtures or extreme quantiles** of the trigger score, used to estimate the direction to project out, did no better than Otsu. The disease signal contaminates any direction estimated from the model's own features.
- **Augmentation during the trigger search** (expectation over transformations) did not make the triggers transfer better or detect better, and made each search several times slower.

9 · Limitations

The artifacts are synthetic and stamped cleanly. Real confounders vary more (several marker styles, partial occlusions, correlations with patient age) and may be learned as distributed features that no single universal edit captures, which is the case where we saw scrubbing fail. Images are 64×64 and the models are small; at clinical resolution, triggers will be harder to search for and to

read. Each condition has two seeds, so differences below about 0.02 AUC are noise. The detection threshold on cost has to be calibrated per task; we show that one shared threshold would fail for chest X-rays. Finally, the real-data audit uses MedMNIST's preprocessed images, which may already have removed the burned-in text and borders through which real shortcuts usually enter.

10 · Conclusion

> A medical classifier's validation score cannot tell you whether it learned the disease or the hospital. Asking the model for its cheapest way to flip a healthy patient to diseased can. On models with planted shortcuts, the answer reproduces the shortcut, including one invisible to people. It identifies most of the affected images, and when the shortcut is global it provides the material to unlearn it. It is not a proof of honesty: honest chest X-ray models are cheap to fool too. The output of an audit is a picture for a person to read, not a pass/fail stamp.

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